

Chapters 1 & 2 Review — Solutions

Full worked solutions to the Chapters 1 & 2 Review Worksheet

How to use this key

- Work every problem first. Only then compare — reading a solution is not the same as producing one.
- Make corrections in **blue pen** next to your original work. Do not erase the mistake; the mistake is the useful part.
- The boxed value is the answer; the lines above it are the reasoning you are expected to show.

Part A — Isolation

§1.1

1. $\frac{2x-1}{3} - \frac{x+4}{5} = 2$

Multiply both sides by 15 (the LCD) to clear fractions: $5(2x - 1) - 3(x + 4) = 30$.

Expand: $10x - 5 - 3x - 12 = 30 \Rightarrow 7x - 17 = 30 \Rightarrow 7x = 47$.

$$\boxed{x = 47/7}$$

2. $\sqrt{3x+1} = x - 3$

Square both sides: $3x + 1 = x^2 - 6x + 9 \Rightarrow x^2 - 9x + 8 = 0 \Rightarrow (x - 1)(x - 8) = 0$.

Candidates $x = 1$ and $x = 8$. Squaring can create false solutions, so check both:

$x = 1$: $\sqrt{4} = 2$ but $x - 3 = -2$. **Extraneous** — the radical is never negative.

$x = 8$: $\sqrt{25} = 5$ and $x - 3 = 5$. Valid.

$$\boxed{x = 8}$$

3. $\frac{3}{x-2} = \frac{5}{x+4}$

Neither denominator may be zero, so $x \neq 2$ and $x \neq -4$. Cross-multiply:

$$3(x + 4) = 5(x - 2) \Rightarrow 3x + 12 = 5x - 10 \Rightarrow 22 = 2x.$$

$$\boxed{x = 11} \quad 11 \text{ is neither excluded value, so it is legitimate.}$$

4. $|2x - 5| = 7$

An absolute value equals 7 when the inside is 7 or -7 , so split into two equations:

$$2x - 5 = 7 \Rightarrow x = 6 \quad \text{or} \quad 2x - 5 = -7 \Rightarrow x = -1.$$

$$\boxed{x = 6 \text{ or } x = -1}$$

5. $\frac{x}{a} + \frac{x}{b} = 1$

Multiply by ab : $bx + ax = ab$. Factor the variable you are solving for: $x(a + b) = ab$.

$$x = \frac{ab}{a + b}$$

This requires $a \neq 0$ and $b \neq 0$ (the original fractions) and $a + b \neq 0$ (the final division). If $a + b = 0$ the equation becomes $0 = ab$, which has no solution for x .

6. $A = P(1 + rt)$, solve for t

Divide by P : $A/P = 1 + rt$. Subtract 1: $rt = A/P - 1 = (A - P)/P$. Divide by r :

$$t = \frac{A - P}{Pr} \quad (P \neq 0, r \neq 0)$$

Part B — Substitution & Elimination

§1.2–1.3

7. $y = 2x - 3$, $3x + 2y = 8$

Substitute the first into the second: $3x + 2(2x - 3) = 8 \Rightarrow 7x - 6 = 8 \Rightarrow x = 2$.

Back-substitute: $y = 2(2) - 3 = 1$. Check in equation 2: $3(2) + 2(1) = 8 \checkmark$

$$(x, y) = (2, 1)$$

8. $4x - 3y = 18$, $3x + 5y = -1$

Eliminate y : multiply the first by 5 and the second by 3 so the y -terms cancel.

$20x - 15y = 90$ and $9x + 15y = -3$. Add: $29x = 87 \Rightarrow x = 3$.

Then $3(3) + 5y = -1 \Rightarrow 5y = -10 \Rightarrow y = -2$. Check: $4(3) - 3(-2) = 18 \checkmark$

$$(x, y) = (3, -2)$$

9. $x + y + z = 6$, $2x - y + 3z = 9$, $x - 2y + 4z = 9$

Eliminate x from the last two using the first. (2) $- 2 \times$ (1): $-3y + z = -3$.

(3) $-$ (1): $-3y + 3z = 3$.

Subtract those two: $2z = 6 \Rightarrow z = 3$. Then $-3y + 3 = -3 \Rightarrow y = 2$, and $x = 6 - 2 - 3 = 1$.

Check in (2): $2(1) - 2 + 3(3) = 9 \checkmark$ and in (3): $1 - 4 + 12 = 9 \checkmark$

$$(x, y, z) = (1, 2, 3)$$

10. $2x - 6y = 5$, $-3x + 9y = 1$

Multiply the first by 3 and the second by 2: $6x - 18y = 15$ and $-6x + 18y = 2$. Add: $0 = 17$, which is false.

No solution

Both equations have slope $1/3$ ($y = x/3 - 5/6$ and $y = x/3 + 1/9$) but different intercepts: the lines are parallel and never meet. A contradiction like $0 = 17$ always means parallel lines; an identity like $0 = 0$ would mean the same line twice.

11. $x^2 + y^2 = 25$, $y = x + 1$

Substitute: $x^2 + (x + 1)^2 = 25 \Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0$.

$x = 3$ gives $y = 4$; $x = -4$ gives $y = -3$. Both satisfy $x^2 + y^2 = 25$.

(3, 4) and (-4, -3)

The circle of radius 5 centered at the origin meets the line $y = x + 1$ in exactly these two points.

12. $x + y = 5$, $y + z = 8$, $z + x = 9$

Add all three: $2(x + y + z) = 22 \Rightarrow x + y + z = 11$.

Subtract each original equation from that total: $z = 11 - 5 = 6$, $x = 11 - 8 = 3$, $y = 11 - 9 = 2$.

(x, y, z) = (3, 2, 6)

Part C — Lines & Graphing

§1.4

13. Line through $(-2, 5)$ and $(4, -7)$

Slope: $m = \frac{-7 - 5}{4 - (-2)} = \frac{-12}{6} = -2$.

Point-slope with $(-2, 5)$: $y - 5 = -2(x + 2) \Rightarrow y = -2x + 1$.

$y = -2x + 1$ y-intercept $(0, 1)$; x-intercept: $0 = -2x + 1 \Rightarrow (1/2, 0)$.

14. Intersection of $2x + 3y = 12$ and $y = x - 1$

Substitute: $2x + 3(x - 1) = 12 \Rightarrow 5x = 15 \Rightarrow x = 3$, so $y = 2$.

(3, 2)

Graph check: $2x + 3y = 12$ has intercepts $(6, 0)$ and $(0, 4)$; $y = x - 1$ passes through $(0, -1)$ and $(1, 0)$. The two lines cross at $(3, 2)$.

15. The line $3x - 4y = 8$

(a) $-4y = -3x + 8 \Rightarrow y = \frac{3}{4}x - 2$, slope $m = 3/4$.

(b) Parallel means equal slope; through $(0, -1)$ the intercept is -1 : **$y = \frac{3}{4}x - 1$**

(c) Perpendicular means the negative reciprocal slope, $-4/3$: **$y = -\frac{4}{3}x - 1$**

16. Ticket problem

Let a = adult tickets and s = student tickets. Then $a + s = 250$ and $12a + 7s = 2300$.
Substitute $s = 250 - a$: $12a + 7(250 - a) = 2300 \Rightarrow 5a + 1750 = 2300 \Rightarrow a = 110$.
So $s = 140$. Check: $12(110) + 7(140) = 1320 + 980 = 2300 \checkmark$

110 adult, 140 student

17. Rectangle with perimeter 54

Let w be the width; then the length is $2w + 3$. Perimeter: $2(w + 2w + 3) = 54$.
 $6w + 6 = 54 \Rightarrow w = 8$, so the length is 19. Check: $2(8 + 19) = 54 \checkmark$

width 8 cm, length 19 cm

Part D — Function Basics

§2.1

18. Domains

(a) $x^2 - 4x - 5 = (x - 5)(x + 1)$, which is 0 at $x = 5$ and $x = -1$.

$(-\infty, -1) \cup (-1, 5) \cup (5, \infty)$ *The numerator's zero at $x = 3$ does not restrict anything.*

(b) Need $7 - 2x \geq 0 \Rightarrow x \leq 7/2$. **$(-\infty, 7/2]$**

(c) The radical needs $x + 4 \geq 0$ and the denominator needs $x + 4 \neq 0$, so $x > -4$. **$(-4, \infty)$**

19. $f(x) = 2x^2 - x + 1$

$f(-3) = 2(9) - (-3) + 1 = 18 + 3 + 1 =$ **22**

$f(a + 1) = 2(a + 1)^2 - (a + 1) + 1 = 2a^2 + 4a + 2 - a - 1 + 1 =$ **$2a^2 + 3a + 2$**

$f(-x) = 2x^2 + x + 1$ — not equal to $f(x)$ or to $-f(x)$, so this function is neither even nor odd.

20. Range of $x^2 - 4x + 7$

$x^2 - 4x + 7 = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$.

A square is never negative, so the output is at least 3, with equality exactly at $x = 2$; every value above 3 is hit as well.

$[3, \infty)$

21. Difference quotient for $f(x) = x^2 - 5x$

$f(x + h) = (x + h)^2 - 5(x + h) = x^2 + 2xh + h^2 - 5x - 5h$.

Subtract $f(x) = x^2 - 5x$: $2xh + h^2 - 5h = h(2x + h - 5)$. Divide by $h \neq 0$:

$2x + h - 5$

22. Which are functions of x ?

(a) Yes — each input 1, 2, 3 appears once. (Two inputs sharing the output 2 is fine.)

(b) No — the input 1 is paired with both 2 and 3.

(c) No — $x = 4$ gives $y = 2$ and $y = -2$; it fails the vertical line test.

(d) Yes — every x produces exactly one y .

Part E — Graphs of Functions

§2.2

23. $y = -2|x - 3| + 4$

Read it from the inside out: shift **right 3**; **stretch vertically** by a factor of 2; **reflect across the x -axis** (the minus sign); shift **up 4**.

Vertex (3, 4), opens downward, with sides of slope -2 and $+2$. Useful check points: (2, 2) and (4, 2); x -intercepts at $x = 1$ and $x = 5$.

24. Transformed points from (2, -5)

(a) $y = f(x - 4) + 1$: input shifts right 4, output up 1 → **(6, -4)**

(b) $y = -3f(x)$: same input, output $\times (-3)$ → **(2, 15)**

(c) $y = f(2x)$: need $2x = 2$, so $x = 1$ → **(1, -5)**

(d) $y = f(-x)$: need $-x = 2$, so $x = -2$ → **(-2 , -5)**

Changes inside the function affect x and act in reverse; changes outside affect y and act as written.

25. $y = \sqrt{x + 2} - 3$

The basic square-root curve shifted left 2 and down 3.

Domain $[-2, \infty)$, range $[-3, \infty)$, endpoint $(-2, -3)$

Plot $(-2, -3)$, $(-1, -2)$, $(2, -1)$, $(7, 0)$.

26. Even, odd, or neither

(a) $f(-x) = -x^3 + 4x = -f(x)$ → **odd** (symmetric about the origin).

(b) $g(-x) = x^4 - 3x^2 + 1 = g(x)$ → **even** (symmetric about the y -axis).

(c) $h(-x) = x^2 - x$, which is neither $h(x)$ nor $-h(x)$ → **neither**.

27. $f(x) = 2x - 1$, $g(x) = x^2 + 3$

$$f(g(x)) = 2(x^2 + 3) - 1 = \boxed{2x^2 + 5}$$

$$g(f(x)) = (2x - 1)^2 + 3 = \boxed{4x^2 - 4x + 4}$$

$$f(g(2)): g(2) = 7, \text{ then } f(7) = \boxed{13}. \quad g(f(2)): f(2) = 3, \text{ then } g(3) = \boxed{12}.$$

Composition is not commutative — order matters.

28. $(f \circ g)(x)$ with $f(x) = 1/(x - 2)$, $g(x) = \sqrt{x}$

$$(f \circ g)(x) = \frac{1}{\sqrt{x} - 2}$$

Two restrictions: x must be in the domain of g , so $x \geq 0$; and $g(x)$ must be in the domain of f , so $\sqrt{x} \neq 2$, i.e. $x \neq 4$.

$$\text{Domain: } [0, 4) \cup (4, \infty)$$

29. $h(x) = (3x + 1)^5$

Take the inside expression as g and the outside operation as f :

$$\boxed{g(x) = 3x + 1 \text{ and } f(x) = x^5} \quad \text{Other splits work too — for instance } g(x) = 3x \text{ with } f(x) = (x + 1)^5 \text{ — but this is the natural one.}$$

30. Repeated composition of $f(x) = 2x - 3$

$$f(f(x)) = 2(2x - 3) - 3 = \boxed{4x - 9}$$

$$f(f(f(x))) = 2(4x - 9) - 3 = \boxed{8x - 21}$$

Four applications give $2(8x - 21) - 3 = 16x - 45$. In general n applications give $2^n x - 3(2^n - 1)$.

31. $f(x) = (2x - 5)/3$

Write $y = (2x - 5)/3$, swap the roles, and solve for the new output: $3y = 2x - 5 \Rightarrow x = (3y + 5)/2$.

$$f^{-1}(x) = \frac{3x + 5}{2}$$

$$\text{Check: } f(f^{-1}(x)) = (2 \cdot \frac{3x+5}{2} - 5)/3 = (3x + 5 - 5)/3 = x \quad \checkmark$$

32. $f(x) = 4 - x^3$

$$y = 4 - x^3 \Rightarrow x^3 = 4 - y \Rightarrow x = \sqrt[3]{4 - y}.$$

$$\boxed{f^{-1}(x) = \sqrt[3]{4 - x}} \quad \text{Domain and range are all real numbers — odd powers are one-to-one, so no restriction is needed.}$$

33. $f(x) = (x + 1)/(x - 3)$

$y(x - 3) = x + 1 \Rightarrow xy - 3y = x + 1$. Collect the x terms: $x(y - 1) = 3y + 1$.

$$f^{-1}(x) = \frac{3x + 1}{x - 1}, \quad x \neq 1$$

The excluded input 1 of f^{-1} is exactly the value f never outputs, and the excluded input 3 of f is the value f^{-1} never outputs. Domain and range swap.

34. $f(x) = x^2 - 6x + 5$ on $x \geq 3$

Complete the square: $f(x) = (x - 3)^2 - 4$. On $x \geq 3$ the function is increasing, so it is one-to-one and invertible there.

$$y = (x - 3)^2 - 4 \Rightarrow (x - 3)^2 = y + 4 \Rightarrow x - 3 = \sqrt{y + 4} \quad (\text{the positive root, since } x - 3 \geq 0).$$

$$f^{-1}(x) = 3 + \sqrt{x + 4}, \quad \text{domain } x \geq -4 \quad (\text{the range of } f \text{ is } [-4, \infty)).$$

35. Short answers

(a) $f^{-1}(8) = 3$ — the inverse undoes f , sending outputs back to inputs.

(b) 5 — $(f^{-1} \circ f)(x) = x$ for every x in the domain of f .

(c) It is not one-to-one: $g(2) = g(-2) = 4$, so an inverse would have to send 4 to two different values. It fails the horizontal line test. (Restricting to $x \geq 0$ fixes this.)

(d) It is the reflection of the graph of f across the line $y = x$; every point (a, b) on f becomes (b, a) on f^{-1} .

Challenge Problems



36. ★ $\sqrt{x + 5} + \sqrt{x} = 5$

Isolate one radical first: $\sqrt{x + 5} = 5 - \sqrt{x}$. Square both sides:

$$x + 5 = 25 - 10\sqrt{x} + x \Rightarrow 10\sqrt{x} = 20 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4.$$

$$\text{Check: } \sqrt{9} + \sqrt{4} = 3 + 2 = 5 \quad \checkmark$$

$$x = 4$$

Isolating before squaring is what keeps the algebra linear; squaring the sum directly leaves a cross term.

37. ★ $f(x) = (2x + 3)/(x - 2)$

$$f(f(x)) = \frac{2 \cdot \frac{2x+3}{x-2} + 3}{\frac{2x+3}{x-2} - 2}. \quad \text{Multiply numerator and denominator by } (x - 2):$$

$$= \frac{2(2x+3) + 3(x-2)}{(2x+3) - 2(x-2)} = \frac{4x + 6 + 3x - 6}{2x + 3 - 2x + 4} = \frac{7x}{7} = x \quad \checkmark$$

$$f \text{ is its own inverse: } f^{-1} = f$$

Its graph is symmetric across the line $y = x$. Such functions are called involutions.

38. ★ $f(x) + 2f(1/x) = 3x$

Substituting $x = 2$: $f(2) + 2f(1/2) = 6$. Substituting $x = 1/2$: $f(1/2) + 2f(2) = 3/2$.

Two equations in the two unknowns $f(2)$ and $f(1/2)$. Double the second and subtract the first: $3f(2) = 3 - 6 = -3$.

$$f(2) = -1$$

In general, replacing x by $1/x$ gives $f(1/x) + 2f(x) = 3/x$. Doubling that and subtracting the original eliminates $f(1/x)$: $3f(x) = 6/x - 3x$.

$$f(x) = \frac{2}{x} - x \quad \text{Check: } f(2) = 1 - 2 = -1 \quad \checkmark$$

39. ★ Range of $f(x) = (2x + 1)/(x - 3)$

Ask which y are achievable: $y = \frac{2x+1}{x-3} \Rightarrow y(x-3) = 2x+1 \Rightarrow x(y-2) = 3y+1$.

If $y \neq 2$ we can solve: $x = (3y+1)/(y-2)$, so that y is attained. If $y = 2$ the equation becomes $0 = 7$, which is impossible.

Range: all real numbers except 2 *2 is the ratio of the leading coefficients — the horizontal asymptote.*